FlowPIE: Test-Time Scientific Idea Evolution with Flow-Guided Literature Exploration

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Research Questions

- How can literature retrieval be a dynamic, adaptive component within idea generation, instead of a static stage?
- How can LLMs leverage retrieved literature and their relationships to generate novel and divergent ideas with continuous refinement?
- Can test-time idea evolution improve scientific idea generation quality beyond static retrieval-then-generation paradigms?

Core Idea

Existing scientific idea generation methods use a static two-stage process: retrieve fixed literature, then generate ideas. This leads to homogeneous ideas trapped in information cocoons. FlowPIE's key insight is to make literature retrieval adaptive by coupling it with idea quality feedback through a flow-guided MCTS inspired by GFlowNets. High-quality generated ideas increase the probability of exploring similar literature paths, creating a dynamic exploration that balances breadth and depth. The framework then evolves ideas using evolutionary operators (crossover, mutation with isolation islands) to incorporate cross-domain knowledge and escape local optima.

Key Formula

Flow-guided UCB

$$UCB(s'|s) = Q(s'|s) + c \cdot P_f(s'|s)\sqrt{\frac{N(s)}{1+N(s'|s)}}$$

UCB formula for node selection in flow-guided MCTS. Q(s'|s) is expected value, P_f(s'|s) is flow probability from GFlowNet inspiration, N(s) and N(s'|s) are visit counts, c is exploration coefficient. Replaces uniform exploration with flow-aware allocation based on idea quality feedback.

Flow probability update

$$P_f(s'|s) \leftarrow (1-\alpha)P_f(s'|s) + \alpha\tilde{R}_t$$

Updates flow probability using exponential moving average. P_f(s'|s) is transition flow probability, α is learning rate, R̃_t is depth-decayed reward. High-reward trajectories accumulate larger flow mass, enabling adaptive literature exploration.

Key Figure

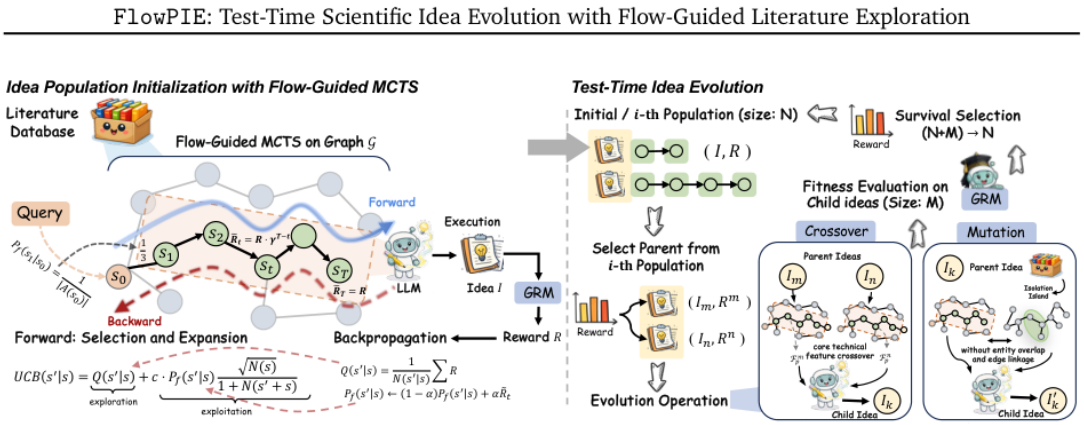

Figure 3 | Overview of FlowPIE. **Left:** Idea initialization based on flow-guided MCTS: forward exploration via flow-guided UCB (Eq.1) for nodes selection and expansion, and backward updating (Eq.2) to enforce local and global flow constraints; this dual process dynamically adjusts literature weights with GRM-based rewards on generated ideas. **Right:** Test-time idea evolution, supervised by GRM-based fitness computation, leveraging various operators, including crossover on core technical features and isolation-island-enhanced mutation, to evolve ideas.

Figure 3: Shows the complete FlowPIE architecture with flow-guided MCTS for idea initialization and evolutionary operators for test-time idea evolution, illustrating the key technical innovation.

Limitations

- Relies on patent database which may not capture all relevant scientific literature compared to academic papers
- Evolutionary process requires multiple LLM calls making it computationally expensive compared to single-shot generation
- Flow-guided MCTS convergence depends on reward variance threshold which may not be optimal across all domains
- GRM-based fitness evaluation introduces additional bias from the reward model's training distribution